

Week 4 — Electric Motors

Sessions: 1 (Theory) · 2 (Lab) · **Total:** 4 hours

Primary LOs: LO-06

Hands-on target: Session 2 ≈ 85% practical

Instructor Preparation Guide (Complete Day Before)

Pre-session setup

- [] Training cart with accessible motor OR bench motor with labeled terminals
- [] Brush length spec sheet for that motor (OEM or training card)
- [] Micrometer/caliper, mirror, flashlight, feeler tools
- [] DMM; megger or high-ohm insulation check method card
- [] Print labs/W04_motor_inspection_report.md
- [] Torque specs for reinstall if students remove motor cover/motor
- [] Confirm F&R contactor location for demo

Planted / selected conditions

Condition	Notes
Worn brushes (near limit)	Ideal for measurement practice
Scored commutator photos if live not available	Use photo set
Motor with elevated no-load current (if safe)	Instructor-supervised dynamic test only

Safety

- Dynamic tests: jack stands/wheel chocks; no bystanders in arc of wheels; pack LOTO when probing internals.
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Week 4 Learning Objectives

1. Identify series, SepEx, and PM/BLDC motor types and operating characteristics.
 2. Perform thorough visual and electrical inspection of DC motors.
 3. Diagnose worn brushes, commutator issues, armature/field faults, bearing failure.
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Session 1 — Theory & Demonstration (2 hours)

Materials needed: Sample motor or cutaway photos; brush samples (new vs worn); F&R contactor for demo; whiteboard PWM sketch; OEM brush-length specs; PPE discussion for rotating parts.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	Safety: rotating parts, high current at A1/A2/S1/S2	
0:05	20	Motor types: Series, SepEx, PM/BLDC	Map to cart generations (Club Car/EZ-GO/ Yamaha examples)
0:25	15	Components: armature, field, commutator, brushes, bearings, fan	[DIAGRAM PLACEHOLDER: series DC motor cutaway with brush gear]
0:40	15	Torque/speed vs controller PWM relationship	Conceptual; no deep control theory
0:55	5	Break	
1:00	15	F&R contactor / reverse logic interaction	Wrong reverse diagnosis often blamed on motor
1:15	20	Wear patterns & service life (brushes 500–2000+ hrs typical range)	Replace criteria: <½ original or OEM spec
1:35	15	Electrical test theory: resistance, shorts to ground	Megger caution on sensitive electronics — isolate motor
1:50	10	Lab plan & lift points review	

Key Teaching Points

- Many “bad motors” are bad cables, solenoids, or controllers — test before condemning.
- Brush dust + moisture → ground faults.
- Commutator mica condition matters; do not file aggressively without training.
- Isolate motor from controller before insulation tests that could damage electronics.

Common Misconceptions

Misconception	Correction
“If it spins by hand it’s electrically fine.”	Opens/shorts may still exist under load.
“Any low ohms reading means good.”	Compare to OEM specs; look for shorts to ground.
“BLDC service is identical to brushed.”	Different diagnostics; often module/sensor related.

Safety Reminders

- Chock wheels; support cart properly before underbody work.
- Pack isolated when fingers are near commutator/brush gear.
- Hot motors after run — burn hazard.

Session 2 — Hands-On Lab (2 hours)

Materials needed: labs/w04_motor_inspection_report.md; training cart/bench motor; calipers; DMM; megger or high-ohm method card; torque specs; wheel chocks; LOTO kit.

Timed Agenda

Time	Min	Activity	Instructor Notes
0:00	5	LOTO verify; PPE; assign motors	
0:05	15	Access motor; visual exterior + cable condition	
0:20	20	Brush length measurement & commutator inspection	Record against spec
0:40	10	Bearing play/noise check	
0:50	5	Break	
0:55	25	Electrical: armature/field resistance; insulation-to-ground	Instructor approves megger use
1:20	20	Dynamic no-load / direction check (supervised)	Record current if clamp available
1:40	15	Complete inspection report + recommend rebuild/replace/OK	
1:55	5	Reassemble critical covers; cleanup	

Assessment Focus

Motor inspection & test report with notes and recommended action.

Week 4 Quiz (8 questions)

1. Separately excited (SepEx) motors differ from series motors primarily in:
 - A) Tire size
 - B) How the field is powered/controlled relative to the armature
 - C) Fuel octane
 - D) Number of solenoids only
2. Brushes are typically replaced when:
 - A) They look shiny

- B) Length is below OEM minimum or $\sim\frac{1}{2}$ original (per training rule/spec)
 - C) The cart is washed
 - D) SG is low
3. A short from winding to motor case is detected by:
- A) Tire pressure
 - B) Insulation resistance / continuity-to-ground checks (power isolated)
 - C) Hydrometer
 - D) Toe plates
4. Forward/Reverse contactors:
- A) Only affect headlights
 - B) Change motor field/armature polarity relationships for direction
 - C) Replace the BMS
 - D) Are never used on Club Car
5. True or False: You should megger a motor while it is still connected to a sensitive controller.

Short Answer

1. List four motor components inspected visually in lab.
2. What does excessive no-load current possibly indicate?
3. Why measure brush length with a real tool instead of “eyeballing”?

Answer Key

1. **B**
2. **B**
3. **B**
4. **B**
5. **False** — isolate first
6. Brushes, commutator, cables/terminals, cooling fan, bearings/housing, mounting, etc.
7. Dragging bearings, partial shorts, incorrect wiring, mechanical bind
8. Specs are quantitative; eyeballing misses marginal wear and creates inconsistent replace decisions